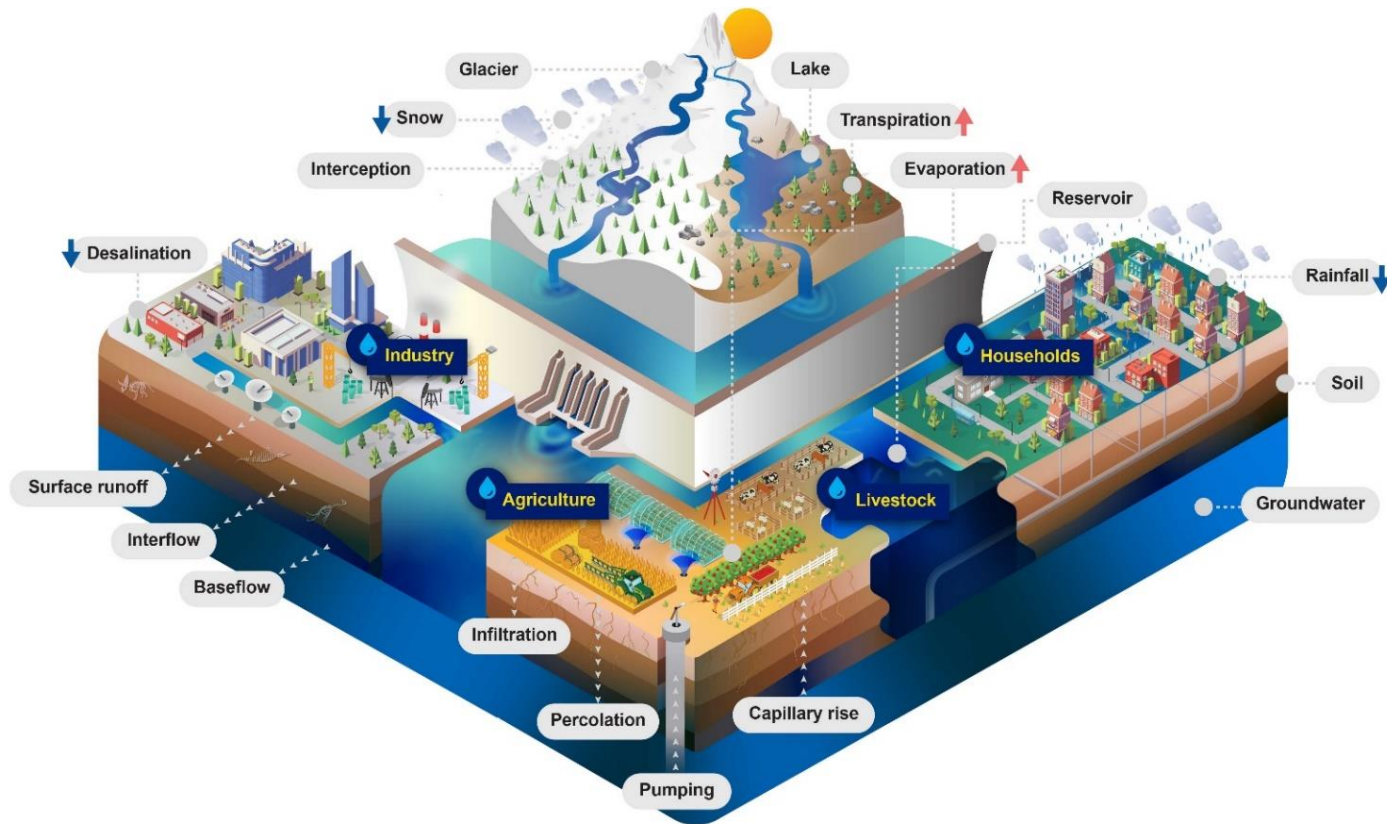




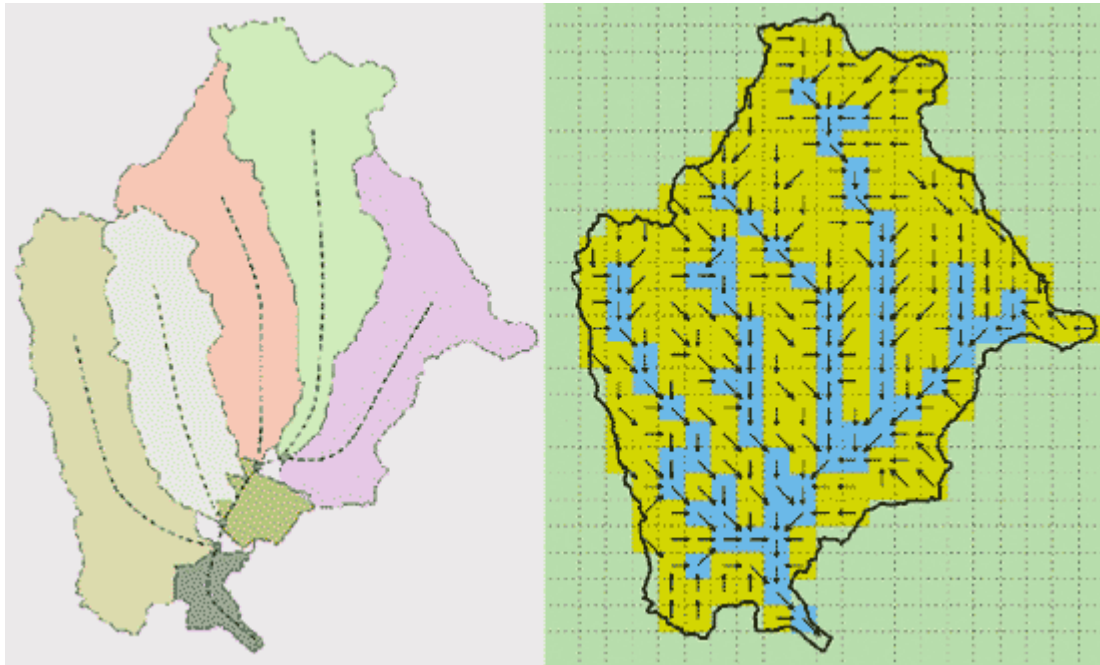
CWatM Tutorial: CWatM Settings file

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Distributed model

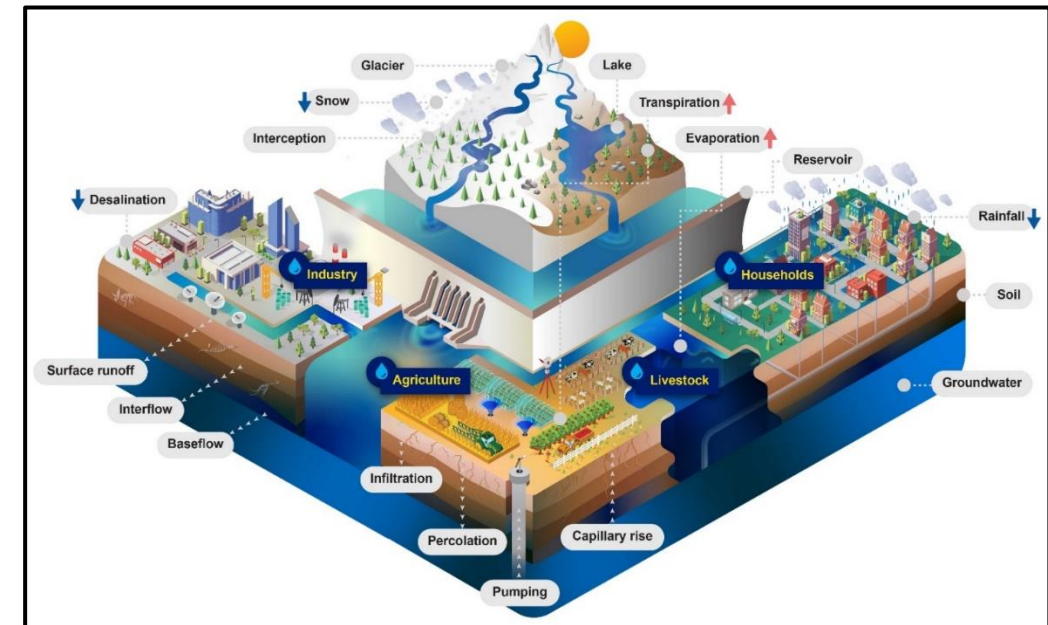
1. Lumped and distributed modeling

- CWatM is a distributed model which can be used at different resolutions e.g. 30 arcmin, 5 Arcmin, 1 min, etc...
- Other hydrological models use a lumped model approach (e.g. SWAT, WEAP)



Lumped model vs. distributed model

(source: <http://crahi.upc.edu>)



Processes – Land cover

The model uses fraction of 6 different land cover types:

- 1) Forest
- 2) Grassland
- 3) Irrigated land
- 4) Paddy Irrigated
- 5) Sealed urban areas
- 6) Water

Processes – Land cover

Pip uninstall numpy

Pip uninstall GDAL

Pip install numpy==1.26

Download (or on a stick) -> GDAL.3.18

Processes – Soil

Three soil layers for each land cover type including processes:

- 1) Frost interrupting soil processes
- 2) Infiltration
- 3) Preferential flow
- 4) Capillary rise
- 5) Surface runoff
- 6) Interflow
- 7) Percolation to groundwater

CWatM settings

Goals:

- 1) Exploring the setting file modules
- 2) Running CWatM using the initialization file in your folder and the command line.
- 3) Generate several output.

First step:

Go to your CWATM folder:

- Open the setting file with a text editor or notepad

```

1  # -----
2
3  ##### ##      ## #### ##### ##  ##
4  ##      ##      ## ## ##  ##  #####
5  ##      ##      ## ## ##  ##  #####
6  ##      ##  ##  ## ##### ##  ##  ##
7  ##      ## #### ##  ##  ##  ##  ##
8  ##      ##### ##  ##  ##  ##  ##
9  ##### ##  ##  ##  ##  ##  ##
10
11 # Community Water Model Version 0.99
12 # SETTINGS FILE
13 # -----
14
15
16 [OPTIONS]
17 #-----
18 # OPTION - to switch on/off
19 #-----
20
21 # Data otions
22 # if temperature is stored in Kelvin instead Celsius
23 TemperatureInKelvin = True
24 # if lat/lon the area has to be user defined = precalculated
25 gridSizeUserDefined = True
26
27 #-----
28 # Evaporation: calculate pot. evaporation (True) or use precalculated pot.evaporation map stacks (False)
29 calc_evaporation = True

```

Folder paths

Most problems can be avoided by setting the folder structure in a confusing way:

We try to set up everything with **relative path**.

1. The settings file has a part: [FILE_PATHS]

PathRoot = ../cwatm_data

\$(PathRoot) is the relative path we will use to access the other input folders, therefore try to organize your folder in a structured way.

2. PathOut = ./output

output in the same folder as the settings file or the directory you are in

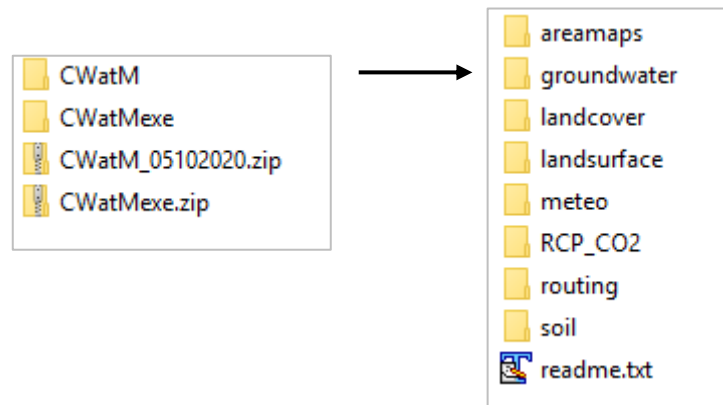
PathMaps = \$(PathRoot)/cwatm_input30min

PathMeteo = \$(PathRoot)/climate/rhine

3.. You can use also absolute path PathRoot =

C:/root/directory/second.root/cwatm/cwatm_data

```
#-----  
[FILE_PATHS]  
#-----  
PathRoot = P:\watmodel\CWATM\cwatm_input_5min  
  
PathOut = P:\watmodel\CWATM\Regions\Danube\output  
PathMaps = $(PathRoot)  
PathMeteo = P:\watxene\ISIMIP\ISIMIP3a\InputData\climate\obsclim_updated\GSWP3-W5E5  
  
#-----  
[NETCDF_ATTRIBUTES]  
institution = IIASA  
title = Rhine Water Model - WATCH WDFEI  
metaNetcdfFile = ../../cwatm//metaNetcdf.xml
```



Time and Initial conditions

- You can run the simulation for the years of interest for you, by changing the StepStart and StepEnd of the file.
- If you have set up a SpinUp period the model will run anyway from 1/1/2010 but it will start to produce the output from 30/1/2010.

```
#-----
[TIME-RELATED_CONSTANTS]
#-----

# StepStart has to be a date e.g. 01/06/1990
# SpinUp or StepEnd either date or numbers
# SpinUp: from this date output is generated (up to this day: warm up)

StepStart = 1/1/2010
SpinUp = 30/1/2010
StepEnd = 10/3/2010

#-----
[INITIAL CONDITIONS]
#-----

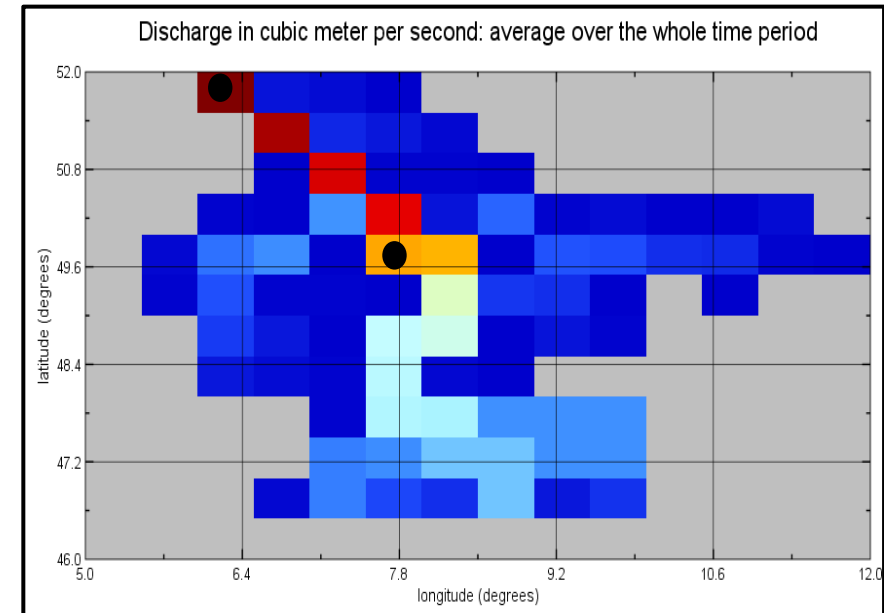
# for a warm start initial variables a loaded
# e.g for a start on 01/01/2010 load variable from 31/12/2009
load_initial = True
initLoad = $(FILE_PATHS:PathRoot)/init/Bhima_20100110.nc

# saving variables from this run, to initiate a warm start next run
# StepInit = saving date, can be more than one: 10/01/1973 20/01/1973
save_initial = True
initSave = $(FILE_PATHS:PathRoot)/init/Bhima
StepInit = 20/1/2010
```


Area & Outlets

- MaskMap can be either a .map file of the catchment of interest or it requires the coordinates of the basin outlet.
- Look for gauges coordinates in the metadata file.
- Change it to the gauges coordinates of interest. You can select single or multiple gauges.

```
47 # -----
48 # AREA AND OUTLETS
49 # -----
50 [MASK_OUTLET]
51
52 # Area mask
53 # A raster map e.g. $(BASICS:PathRoot)\data\areamaps\area_indus.map
54 # or a rectangle: Number of Cols, Number of rows, cellsize, upper left corner X, upper left corner Y
55
56 MaskMap = $(BASICS:PathRoot)\data\areamaps\area3.map
57 # Indus
58 #MaskMap = 30 20 0.5 65 38
59 #MaskMap = $(BASICS:PathRoot)\data\areamaps\area_indus.map ; Cut out Indus only
60 # Rhine
61 #MaskMap = 30 20 0.5 3 54
62
63 # Station data
64 # either a map e.g. $(BASICS:PathRoot)\data\areamaps\area3.map
65 # or a location coordinates (X,Y) e.g. 5.75 52.25 9.25 49.75 )
66
67 #Gauges = $(BASICS:PathRoot)\data\areamaps\station8.map
68 # Rhine
69 Gauges = 5.75 52.25 9.25 49.75
70
71 # -----
72 [TIME-RELATED_CONSTANTS]
73 # -----
74
75 # StepStart and Stepend either dates e.g. 01/06/1990
```



Output

1. Go to the last lines of your setting file
2. Different format for your output: .tss and .nc
3. You can comment the output you do not need for your analysis in order to save computational time

```
#+++++

[OUTPUT]

# OUTPUT maps and timeseries
OUT_Dir = $(FILE_PATHS:PathOut)

OUT_TSS_Daily = discharge
#OUT_TSS_MonthAvg = discharge
#OUT_TSS_AnnualAvg = discharge

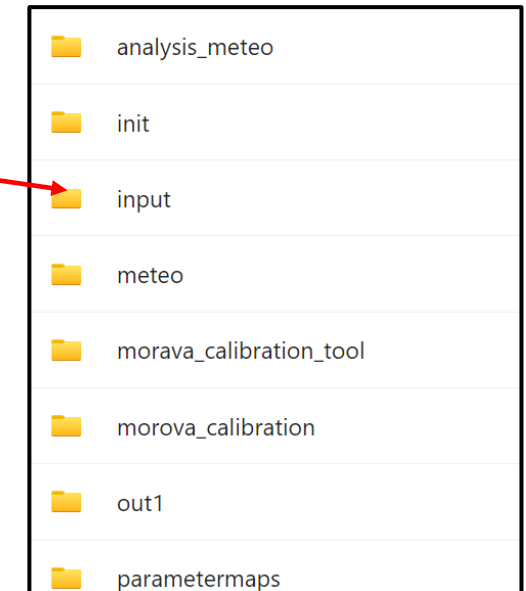
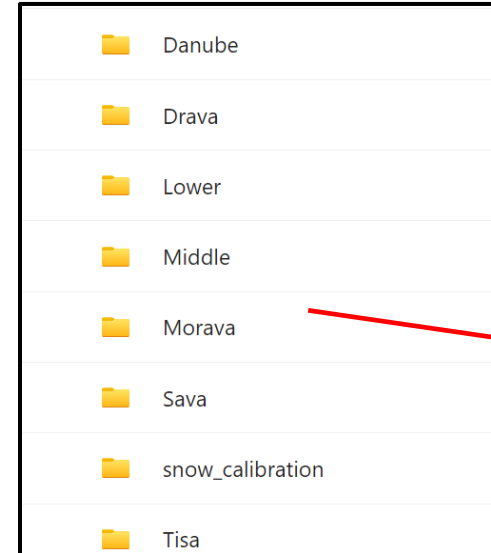
#OUT_Map_Daily = discharge
#OUT_Map_MonthAvg = discharge, precipitation, runoff
#OUT_Map_AnnualAvg = discharge
#OUT_MAP_TotalAvg = discharge, baseflow

OUT_MAP_Daily = Rain, Snow, act_nonpaddyConsumption, act_paddyConsumption, act_livConsumption, returnflowIrr,
returnflowNonIrr, GW_Irrigation, Res_Irrigation, Lake_Irrigation, Channel_Irrigation, GW_Industry, GW_Livestock,
GW_Domestic, Res_Industry, Res_Livestock, Res_Domestic, Lake_Industry, Lake_Livestock, Lake_Domestic,
Channel_Domestic, Channel_Livestock, Channel_Industry, preflow_GW, perc3toGW_GW, lakeResInflowM, lakeResOutflowM,
act_indConsumption, act_domConsumption, act_livConsumption, act_irrWithdrawal, act_nonIrrWithdrawal,
act_domWithdrawal, act_indWithdrawal, act_livWithdrawal, snowEvap, capillar, baseflow, actTransTotal_forest,
actTransTotal_grasslands, actTransTotal_paddy, actTransTotal_nonpaddy, unmet_lost, unmetDemand,
pot_GroundwaterAbstract, storGroundwater, discharge, nonFossilGroundwaterAbs, Precipitation, totalET,
EvapoChannel, EvapWaterBodyM, act_nonIrrConsumption, channelStorage, lakeResStorage, totalSto, sum_actTransTotal,
sum_actBareSoilEvap, sum_interceptEvap, sum_openWaterEvap, addtoevapotrans, lakeResInflowM, act_bigLakeResAbst,
sum_gwRecharge, sum_capRiseFromGW, act_totalIrrConsumption, sum_runoff, returnFlow, act_SurfaceWaterAbstract,
dis_outlet, act_irrNonpaddyWithdrawal
OUT_MAP_TotalEnd = cellArea
```

Hands on CWatM 1 min – Exercise

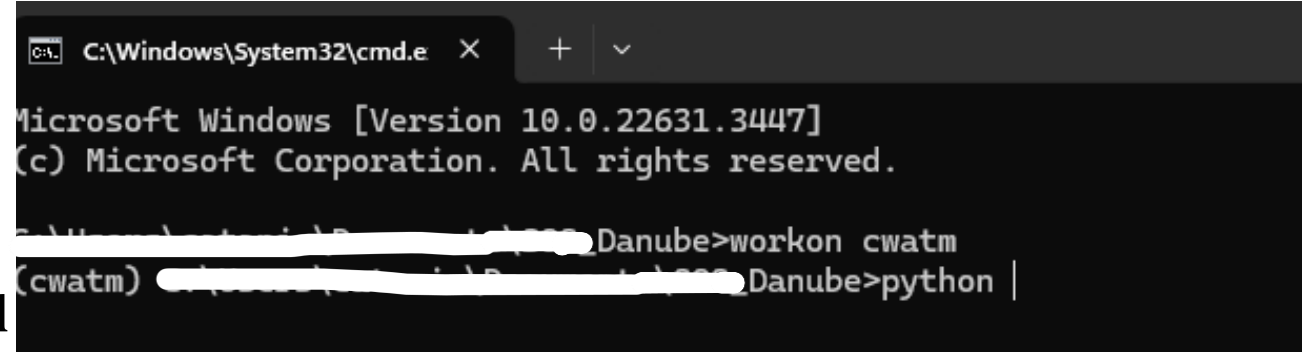
Play around with your own catchment.

- Change the settings file:
 - Run for different times
 - Produce different outputs
- What catchment are you interested?
 - Find out the coordinates (lat/lon) of the outlet point
 - Find out coordinates of gauges
 - Send me the coordinates for next lesson



Hands on CWatM – Exercise

- You can execute CWatM opening the command line from the folder where you have your setting file
- Activate your Virtual Environment
- (venv)C/User/ ...folder_path> python C/Github_folderpath/run_cwatm.py setting_file -l



```

C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.22631.3447]
(c) Microsoft Corporation. All rights reserved.

C:\Users\...>workon cwatm
(cwatm) C:\Users\...>python |
  
```